

Human-Scale Personal Fabrication

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ABSTRACT

Building large structures from small elements, creating life-size animated creatures, or making contraptions that we can ride on have almost certainly been everyone's childhood dreams. However, researchers and practitioners of personal fabrication have been mainly focusing on creating objects that fit into a human palm, also called "hand-size" objects. The reason behind this is not only because of the size limitation of consumer-grade fabrication machinery but also because of the very long printing time and high material costs of large-scale prototypes. To overcome these limitations, I combine 3D printed hubs and ready-made objects, such as plastic bottles, as well as welding steel rods into a certain type of node-link structures called "trusses". However, the actual challenge behind my work is not only about achieving the size, but ensuring that the resulting large structures withstand the orders of magnitude larger forces than their hand-sized counterparts. Designing such structures requires substantial engineering know-how that users of personal fabrication equipment, such as makers, typically do not possess. By providing the lacking engineering know-how, my three end-to-end software systems TrussFab, TrussFormer, and Trusscillator enable non-experts to build such human-scale static, kinetic, and human-powered dynamic devices, such as pavilions, large-scale animatronic devices, and playground equipment. These systems achieve this by allowing users to focus on high-level design aspects, such as shape, animation, or riding experience, while abstracting away the underlying technicalities of forces, inertia, eigenfrequencies, etc. To help building the designs, the software generates the connector pieces and assembly instructions. With this body of work, I aim at democratizing engineering that enables individuals to design and fabricate large-scale objects and mechanisms that involve human-scale forces.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools..

KEYWORDS

personal fabrication, 3D printing, CAD design, force-aware fabrication;

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1 INTRODUCTION

Personal fabrication tools, such as 3D printers have become popular in HCI, where they have been used for fast prototyping [12], as well as to fabricate interactive soft objects [6], or kinematic characters [5]. Since 3D printers today are available in a desktop form factor, they have been able to spread to the maker community and are now increasingly reaching the consumer market.

In contrast, the fabrication of large-scale objects still has remained a privilege of industry, which has access to specialized equipment, such as concrete printers that allow making houses [8] or robotic-arms capable of 3D printing [7]. The owners of the widespread desktop devices cannot participate in this evolution, because the underlying technology does not scale. Even techniques that break down large models into printer-sized parts [13] quickly become intractable, as with scaling the size the volume of an object grows cubed, therefore the material use and printing time increase similarly. This renders 3D printing and related techniques unattractive for larger-than-desktop-scale models.

My first attempt to overcome this size limitation was by creating a hand-held device, called Protopiper [1], that enables the physical prototyping of room-sized objects. While this approach handles the issue of long printing time and material use, it quickly reaches its limitations, since the created objects can only serve a visual purpose, but cannot withstand any external load.

Ultimately, this problem led me to my actual research agenda: The main challenge when creating large objects is not only overcoming the size limitation but to design for orders of magnitude *larger forces* than expected by hand-size objects. This criterion becomes even more challenging considering that the load-bearing properties of materials do not scale proportionally with their size, but typically there is a nonlinear relationship between size, shape and load-bearing capabilities. Therefore, designing for large forces requires substantial *engineering skills* from envisioning the appropriate structure in the first place to verifying their structural integrity.

To investigate this paradigm, I have developed three fabrication systems, called TrussFab [9], TrussFormer [10], and Trusscillator [11]. They consequently dig deeper in assisting users with engineering know-how along the three pillars of mechanical engineering: statics, kinematics, and dynamics. The division of these projects is illustrated in Figure 1.

On a broader scale, embedding *engineering domain knowledge* into consumer software (in my case as a plugin for SketchUp [14]) is a key step to transition large-scale fabrication from the industry to the wider audience, aka. personal fabrication, as identified by Baudisch and Mueller [2].

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movement	TrussFormer (kinematics)	Trusscillator (dynamics)
no movement	Protopiper (shape - visual only)	TrussFab (statics)
	no force	force

Figure 1: Division of my research projects by their supported mechanical function and the covered field of mechanical engineering.

In the following, I give a brief overview of the three mentioned systems, highlighting their specific engineering domain-specific aspects.

2 STATICS – TRUSSFAB

TrussFab [9] is an integrated end-to-end system that allows users to fabricate large-scale static structures that are sturdy enough to carry human weight. TrussFab achieves this by affording truss-based design and fabricates the objects using a desktop 3D printer and plastic bottles.

The key idea behind TrussFab is to afford sturdy “closed frame structures”, also known as trusses [4]. While bending would break easily the 3D printed nodes, truss structures consist of triangles-only configurations that prevent bending at the nodes and redirects the forces into compression and tension, that the bottles are great at handling. TrussFab’s editor (Figure 2a) affords to build trusses by combining tetrahedra and octahedra truss primitives by stacking as voxels. As a result, everything created using these primitives remains a rigid truss. Additionally, it performs structural analysis and assists the fabrication by generating the connector pieces and assembly instructions (Figure 2b).

3 KINEMATICS – TRUSSFORMER

TrussFormer [10] takes a step forward in the direction of moving objects, still building on the principle of trusses. It helps to design large-scale kinetic structures, i.e., structures that involve motion and are subject to inertial forces, such as animatronics devices.

TrussFormer enables movement in a normally rigid truss structure by replacing TrussFab’s formerly rigid hubs with a so-called spherical-hinge mechanism. As shown in Figure 2a, users only need to demonstrate the intended movement on their rigid T-Rex-shaped truss by dragging it towards the desired direction; while TrussFormer in turn picks the best fitting location for a linear actuator and places the hinges at the affected nodes.

After defining the animation pattern that moves the T-Rex body up and down, TrussFormer computes the *inertial forces*, as shown in Figure 2b. Here, the large acceleration of the long neck leads to very high inertial forces, exceeding the breaking limit of the construction. These situations are hard to foresee because the inertial forces can be multiple of the static load in the structure. TrussFormer addresses this by automatically correcting the animation sequence by either limiting the acceleration slope or the range of the movement, assuring that the structure will now withstand the movement.

4 DYNAMICS – TRUSSCILLATOR

Trusscillator [11] goes beyond only fulfilling the criteria of structural stability. It also predicts the dynamic behavior of spring-based oscillating mechanisms, helping to design the riding experience of the device.

To allow designers to create human-powered movement, Trusscillator offers a novel set of tools, specifically designed for dynamic experiences (Figure 4a). These tools allow designers to focus on user experience-specific aspects, such as *motion range*, *tempo*, and *effort* while abstracting away the underlying technicalities of eigenfrequencies, spring constants, masses, and energy use.

Trusscillator achieves this functionality by incorporating a differential equation solver that accurately simulates the dynamics of spring-damper-rod networks, implemented in *Julia* programming language [3].

Since the forces involved in the resulting devices can be high, Trusscillator’s structures are built out of steel trusses. Trusscillator helps users fabricate these not only by picking out appropriate springs but also by producing stencils, placing temporary connectors that help weld the resulting large-scale structures.



Figure 2: (a) TrussFab’s editor affords truss-based design using primitives. Here the user is invoking the integrated structural analysis to verify if the structure will hold up its weight. (b) After the design has been verified, it generates the 3D printable connector pieces, that (c) assemble into a pavilion using ~1200 plastic bottles, holding its own weight (~250 kg).

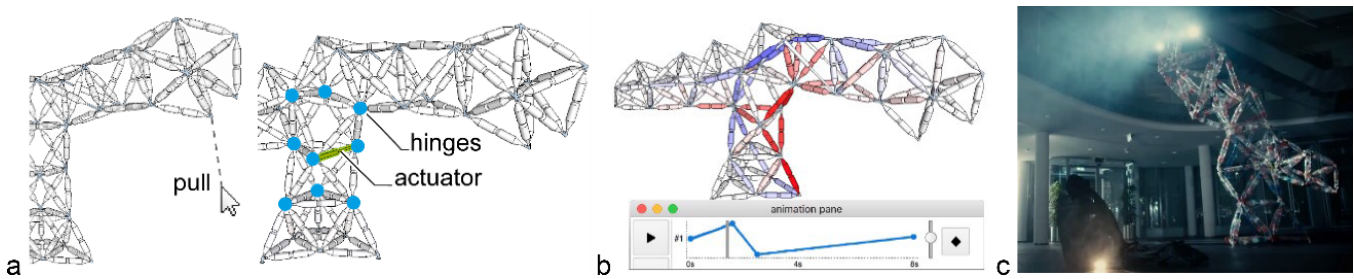


Figure 3: (a) TrussFormer enables movement in the truss structure defined by demonstration. (b) The system assists users in creating animation sequences that won't break the structure because of high inertial forces. (c) The physically built T-Rex model performs the designed motion.

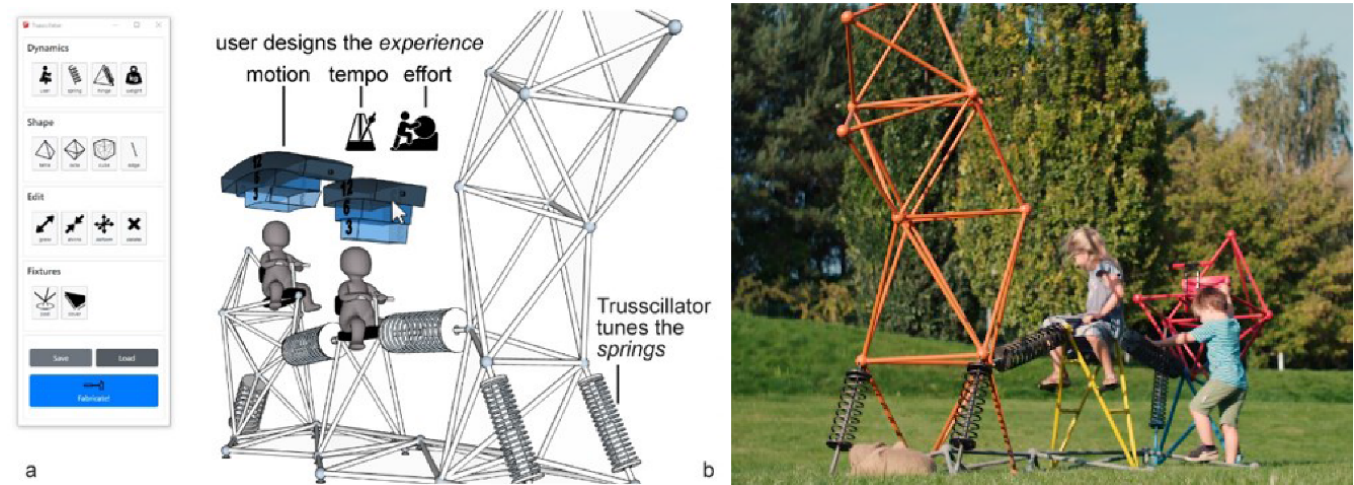


Figure 4: (a) Trusscillator allows designers to add coil springs to their truss-based design and specify the requirements in terms of motion range, speed, and physical effort. (b) A brachiosaurus-swing designed for a comfortable ride for 3 to 6 years old.

5 CONCLUSION

The presented systems aim at solving three aspects of personal fabrication of human-scale objects: statics, kinematics, and dynamics. With this body of work, I aim at emphasizing the necessary steps towards democratizing engineering that enable individuals to design and fabricate large-scale objects and mechanisms that involve human-scale forces.

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